

Perspective: The One-Body Green's Function in Physics and Chemistry

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Abstract: abstract abstract ...

I. INTRODUCTION

Although electronic structure theory has made significant progress in the last decades, there are still phenomena that can not routinely be described with sufficient accuracy using the available methods. For example, strongly correlated materials with different competing electronic phases require a sufficient accuracy in order to correctly disentangle them. At the same time, finite size effects have to be critically assessed, requiring computationally tractable yet accurate methods. Similarly, metal clusters in biological systems, or the description of certain type of excitations, e.g., double excitations, are not routinely captured with sufficient accuracy. Although progress has been made in these areas, significant limitations remain, highlighting the need for new theoretical and computational approaches that combine accuracy with computational efficiency. One promising direction is the development of novel diagrammatic methods based on the one-particle Green's function ($\mathbf{G}$), which are currently being explored by several groups through interdisciplinary efforts.

The one-particle Green's function offers a particularly insightful bottom-up perspective on electronic structure theory. By nature, electrons are indistinguishable and obey the Pauli exclusion principle, which fundamentally shapes their collective behavior in multi-electronic systems. Nevertheless, understanding the collective behavior of a system through its changes in the presence of an additional electron or hole is useful conceptually. From this behavior, information about the ground and excited states of a system can be inferred, and this is what is captured in the one-particle Green's function. Imagine injecting a hole in a system; depending on how strongly the electrons interact with this hole, one can deduce information about the electronic correlations present in the system. If the response is weak, the hole moves relatively freely through the system, essentially behaving as an independent particle. Conversely, if the response is strong, the hole becomes entangled with the surrounding electrons, which ultimately could result in a loss of its single-particle character. The dressed electron/hole is often referred to as a quasiparticle.

From a theoretical point of view, the central ingredient one needs to include for a quasiparticle description is the effective interactions of the electron/hole with its environment. The self-energy, often denoted as $\Sigma(\mathbf{r}, \mathbf{r}'; \omega)$, encapsulates these ef-

fective non-local and generally energy-dependent interactions. Interestingly, the simplest approximation to $\Sigma(\mathbf{r}, \mathbf{r}'; \omega)$ is the Hartree-Fock (HF) approximation in which the self-energy is energy-independent and corresponds to the Coulomb and exchange Fock matrix contributions. In this case, one describes a particle interacting with the averaged field, as in regular HF theory. The central equation that connects the single-particle and the quasiparticle picture is the Dyson equation

$$\mathbf{G}(\omega) = \mathbf{G}_0(\omega) + \mathbf{G}_0(\omega)\mathbf{\Sigma}(\omega)\mathbf{G}(\omega), \quad (1)$$

with $\mathbf{G}_0(\omega)$ denoting some initial independent-particle Green's function that becomes transformed into an interacting Green's function by the interactions described by the self-energy ($\mathbf{\Sigma}$). From this, the importance of infinite resummation becomes evident. Only when the infinite series is summed to all orders does the Green's function fully capture the effects of interactions and provide an accurate description of the quasiparticle properties.

Alternatively, $\mathbf{G}$ can be constructed directly from the many-body wavefunctions of the ground- (Ψ_0) and excited states ($\Psi_m^{N\pm 1}$)

$$G_{pq}(\omega) = \sum_m^{N+1} \frac{\langle \Psi_0 | a_p | \Psi_m^{N+1} \rangle \langle \Psi_m^{N+1} | a_q^\dagger | \Psi_0 \rangle}{\omega - (E_m^{N+1} - E_0) + i\eta} + \sum_m^{N-1} \frac{\langle \Psi_0 | a_q^\dagger | \Psi_m^{N-1} \rangle \langle \Psi_m^{N-1} | a_p | \Psi_0 \rangle}{\omega - (E_0 - E_m^{N-1}) - i\eta}. \quad (2)$$

Formally, both approaches are exact and constitute a different philosophy. While the Dyson equation [Eq. (1)] builds $\mathbf{G}$ without knowledge of the explicit ground-state of the system through, the wavefunction-based construction requires explicit knowledge of the ground-state and excited states of the system [Eq. (2)]. The latter procedure is often denoted as non-Dyson Green's function construction.

Either way, it becomes clear that $\mathbf{G}$ provides direct access to the ionization potentials and electron affinities of the system, which allow for direct comparison with (inverse) angle-resolved photoemission spectroscopy [AR(I)PES] experiments through the spectral function $A(\omega)$, defined as

$$A(\omega) = -\frac{1}{\pi} \text{Im Tr } \mathbf{G}(\omega). \quad (3)$$

Furthermore, the ground-state energy and density matrix can

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be deduced from the Green's function as

$$E_0 = \frac{1}{4\pi i} \sum_{pq} \oint_{\text{IP}} d\omega (h_{pq} + \omega \delta_{pq}) G_{qp}(\omega) \quad (4)$$

$$\gamma_{pq} = \frac{1}{2\pi i} \oint_{\text{IP}} d\omega G_{qp}(\omega), \quad (5)$$

where $\mathbf{h}$ is the one-electron Hamiltonian, γ is the one-particle density matrix, and $\oint_{\text{IP}}$ denotes a contour integral in the upper half of the complex plane.

With this background, we aim to highlight recent advances in constructing approximations to the one-particle Green's function and their implications for electronic structure theory. Along the way, we identify open research questions that must be addressed to advance our understanding of the challenges outlined at the outset.

Maybe provide overview figure of the different techniques.

As its origin in quantum field theory suggests, common strategies for finding self-energy approximations rely on diagrammatic expansions and perturbative techniques. The central problem lies in the inclusion of many-body effects in the self-energy, which describe the adjustment of the electronic structure due to the additional electron or hole, i.e., the dressing through interactions with the surrounding electrons. The most famous example of such a procedure is the *GW* approximation. Here the electron/hole is dynamically dressed by a polarization cloud of (collective) electron-hole pairs. The *GW* approximation constitutes the simplest approximation to a more general complicated self-consistent many-body procedure introduced by Lars Hedin. Following this, the recipe would allow for the construction of the exact self-energy. Such an approach is overly complex and computationally demanding for practical applications. Finding practical ways to go beyond *GW* while not solving the full set of Hedin's equations is a significant challenge.

Alternative theoretical strategies rely on putting a particular emphasis on effective two- (or higher-order) particle interactions that are then used to construct approximations to the self-energy. A famous example is the Parquet approximation. Here different two particle scattering channels are treated on an equal footing, and their effect is incorporated in the self-energy through the Dyson-Schwinger equation. While offering an attractive route also to understand microscopic effects that give rise to electronic phases in correlated condensed matter, the Parquet approximation suffers from similar problems as Hedin's equation when going beyond the lowest order, and a systematically improvable route remains challenging.

Alternatively, one can make use of the

II. ONE-PARTICLE GREEN'S FUNCTION

As a sum-over-state expression for the interacting one-particle Green's function reads

$$G_{pq}(\omega) = \sum_N \langle \Psi_0 | a_p | \Psi_N \rangle (\omega - \Delta E_N^+)^{-1} \langle \Psi_N | a_q^\dagger | \Psi_0 \rangle + \sum_M \langle \Psi_0 | a_q^\dagger | \Psi_M \rangle (\omega - \Delta E_M^-)^{-1} \langle \Psi_M | a_p | \Psi_0 \rangle \quad (6)$$

with $a_q^\dagger$ and a_p denoting fermionic creation and annihilation operators, and $\Delta E_N^+ = E_N^+ - E_0$ and $\Delta E_M^- = E_0 - E_M^-$. With + and - referring to the energy of system with an added/removed electron.

As an alternative to Eq. (6), the one-particle Green's function can be determined through the Dyson equation

$$G_{pq}(\omega) = G_{pq}^0(\omega) + \sum_{rs} G_{pr}^0(\omega) \Sigma_{rs}(\omega) G_{sq}(\omega), \quad (7)$$

with $G_{pq}^0(\omega)$ denoting the non-interacting Green's function

$$G_{pq}^0(\omega) = \delta_{pq} \frac{1 - n_p}{\omega + \epsilon_p} + \delta_{pq} \frac{n_p}{\omega - \epsilon_p}, \quad (8)$$

with n_p denoting the occupation number from the Fermi-Dirac distribution. We assume the zero temperature limit throughout, n_p associated with a mean-field orbital p is therefore either 0 (for an unoccupied orbital) or 1 (for an occupied orbital). Eq. (7) can also be rearranged to (in explicit matrix form)

$$\mathbf{G}(\omega) = \frac{1}{\mathbf{G}_0^{-1}(\omega) - \mathbf{\Sigma}(\omega)}. \quad (9)$$

The connection between the sum-over-state expression and the Dyson equation is made explicit in the following. For this, we define a new vacuum $|\Psi_0\rangle$

$$|\Psi_0\rangle = e^\sigma |0\rangle, \quad (10)$$

via exponential parametrization of the reference mean-field state $|0\rangle$. σ denotes an anti-Hermitian particle-hole excitation operator

$$\sigma = \hat{T} - \hat{T}^\dagger, \quad (11)$$

with

$$\hat{T} = \sum_\nu t_\nu \bar{\tau}_\nu. \quad (12)$$

$\bar{\tau}_\nu$ represents an arbitrary particle-hole excitation operator ($\bar{a}_a^\dagger \bar{a}_i + \dots$). Note that the amplitudes t_ν [Eq. (12)] are determined by projection and satisfy the general Brioullin conditions

$$\langle \Psi_0 | [\hat{H}, \bar{\tau}_\nu - \bar{\tau}_\nu^\dagger] | \Psi_0 \rangle = 0. \quad (13)$$

In the following, occupied orbitals are referred to with subscripts $i, j, k, \dots$, virtual orbitals with $a, b, c, \dots$ and arbitrary orbitals with $p, q, r, \dots$. Furthermore, the particle and hole

creation and annihilation operators acting on the vacuum are (still obeying their anti-commutation relations)

$$\bar{a}_p = e^\sigma a_p e^{-\sigma} \quad (14)$$

$$\bar{a}_p^\dagger = e^\sigma a_p^\dagger e^{-\sigma}. \quad (15)$$

Within this formalism an arbitrary excited state N is obtained as

$$|\Psi_N\rangle = \hat{R}_N |\Psi_0\rangle. \quad (16)$$

$\hat{R}_N$ is given as

$$\begin{aligned} \hat{R}_N |\Psi_0\rangle &= \sum_\nu R_\nu^N \tilde{\tau}_\nu |\Psi_0\rangle \\ &= \sum_\nu R_\nu^N e^\sigma \hat{\tau}_\nu e^{-\sigma} |\Psi_0\rangle = \sum_\nu R_\nu^N e^\sigma |\Phi_\nu\rangle, \end{aligned} \quad (17)$$

with R_ν^N denoting the expansion coefficients so that

$$|\Psi_N\rangle = \sum_\nu R_\nu^N e^\sigma \hat{\tau}_\nu |0\rangle. \quad (18)$$

Note that $\hat{\tau}_\nu$ is chosen according to the excited state of interest, e.g., for charged ($N \pm 1$) excitations

$$\hat{\tau}_\nu^- = \{a_i, a_b^\dagger a_j a_i, \dots\} \quad (19)$$

$$\hat{\tau}_\nu^+ = \{a_a^\dagger, a_a^\dagger a_b^\dagger a_j, \dots\} \quad (20)$$

$$\hat{\tau}_\nu = \{\hat{\tau}_\nu^-\} \cup \{\hat{\tau}_\nu^{+, \dagger}\} \quad (21)$$

so that

$$\begin{aligned} 1 &= \sum_\nu \tilde{\tau}_\nu |\Psi_0\rangle \langle \Psi_0 | \tilde{\tau}_\nu^\dagger \\ 1 &= \sum_\nu e^\sigma \hat{\tau}_\nu |0\rangle \langle 0 | \hat{\tau}_\nu^\dagger e^{-\sigma} \end{aligned} \quad (22)$$

is satisfied. The Hamiltonian for that new vacuum is defined in terms of the new set of creation and annihilation operators

$$\begin{aligned} \bar{H} &= e^\sigma \hat{H} e^{-\sigma} \\ &= \sum_{pq} F_{pq} \bar{a}_p^\dagger \bar{a}_q + \frac{1}{4} \sum_{pqrs} V_{pqrs} \bar{a}_p^\dagger \bar{a}_q^\dagger \bar{a}_r \bar{a}_s. \end{aligned} \quad (23)$$

Within the newly defined quantities excitation energy differences ΔE_N are evaluated using the EOM approach

$$\begin{aligned} &\langle \Psi_0 | [\tilde{\tau}_\nu^\dagger, [\bar{H}, \sum_\mu R_\mu^N \tilde{\tau}_\mu]] | \Psi_0 \rangle \\ &= \Delta E_N \langle \Psi_0 | \tilde{\tau}_\nu^\dagger, \sum_\mu R_\mu^N \tilde{\tau}_\mu | \Psi_0 \rangle, \end{aligned} \quad (24)$$

which can be rewritten to

$$\begin{aligned} &\langle 0 | [\hat{\tau}_\nu^\dagger, [\hat{H}, \sum_\mu R_\mu^N \hat{\tau}_\mu]] | 0 \rangle \\ &= \langle 0 | [\hat{\tau}_\nu^\dagger, \sum_\mu R_\mu^N \hat{\tau}_\mu] | 0 \rangle \Delta E_N. \end{aligned} \quad (25)$$

Eq. (25) resembles the common EOM configuration interaction approach and can be explicitly written in matrix form

$$\mathbf{H}\mathbf{R} = \mathbf{S}\mathbf{R}\mathbf{E} \quad (26)$$

$$\Leftrightarrow \tilde{\mathbf{H}}\mathbf{R} = \mathbf{R}\mathbf{E} \quad (27)$$

(assuming $\mathbf{S}$ is invertible and a symmetric transformation). With the above definitions, Eq. (6) reads

$$G_{pq}(\omega) = \sum_N \langle \Psi_0 | \tilde{a}_p | \Psi_N \rangle (\omega - \Delta E_N^+)^{-1} \langle \Psi_N | \tilde{a}_q^\dagger | \Psi_0 \rangle + \sum_M \langle \Psi_0 | \tilde{a}_q^\dagger | \Psi_M \rangle (\omega - \Delta E_M^-)^{-1} \langle \Psi_M | \tilde{a}_p | \Psi_0 \rangle, \quad (28)$$

which can be rearranged to

$$\mathbf{G}_{11}(\omega) = \begin{pmatrix} \mathbf{1}_{11} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}_{11} \\ \mathbf{0} \end{pmatrix}, \quad (29)$$

where the subscript 11 denotes the p/h subspace and 2 refers to all higher order excitations, i.e., hhp, pph etc.. The connection to the Dyson equation becomes apparent by rewriting Eq. (29)

$$\begin{aligned} \mathbf{G}_{11}(\omega) &= \left[\omega \mathbf{1}_{11} - \mathbf{H}_{11} - \mathbf{H}_{12} [\omega \mathbf{1}_{22} - \mathbf{H}_{22}]^{-1} \mathbf{H}_{21} \right]^{-1} \\ \Rightarrow \mathbf{G}(\omega) &= \left[\omega - \mathbf{F} - \Sigma(\infty) - \tilde{\Sigma}(\omega) \right]^{-1} \\ \Leftrightarrow \mathbf{G}(\omega) &= [\omega - \mathbf{F} - \Sigma(\omega)]^{-1} \\ \Leftrightarrow \mathbf{G}(\omega) &= \left[\mathbf{G}_0^{-1}(\omega) - \Sigma(\omega) \right]^{-1}. \end{aligned} \quad (30)$$

Consequently, we have demonstrated the connection of the effective Hamiltonian used for the determination of single-particle excitation energies and the Dyson equation of the one-particle Green's function. For generality, we have partitioned H_{11} into the bare Fock matrix and additional screening effects summarized in $\Sigma(\infty)$ (for example HF exchange is often considered to be part of $\Sigma(\infty)$). Furthermore, particle and hole space are now intrinsically coupled in $\Sigma(\omega)$.

The route described above is not the strategy commonly used in quantum chemistry for the determination of the one-particle Green's function. Instead, the original definition of creation and annihilation operators is kept, so that the EOM reads

$$\begin{aligned} &\langle 0 | [\hat{r}_\nu^\dagger, [\tilde{H}, \sum_\mu R_\mu^N \hat{r}_\mu]] | 0 \rangle \\ &= \langle 0 | [\hat{r}_\nu^\dagger, \sum_\mu R_\mu^N \hat{r}_\mu] | 0 \rangle \Delta E_N, \end{aligned} \quad (31)$$

with $\tilde{H}$ denoting the unitary transformed Hamiltonian

$$\tilde{H} = e^{-\sigma} \hat{H} e^\sigma \quad (32)$$

and the one-particle Green's function written as

$$\begin{aligned} G_{pq}(\omega) &= \sum_{\nu, \mu} \langle 0 | e^{-\sigma} a_p e^\sigma | \Phi_\nu \rangle \left[(\omega - \mathbf{H}^+)^{-1} \right]_{\nu\mu} \langle \Phi_\mu | e^{-\sigma} a_q^\dagger e^\sigma | \Psi_0 \rangle \\ &+ \sum_{\nu, \mu} \langle 0 | e^{-\sigma} a_q^\dagger e^\sigma | \Phi_\nu \rangle \left[(\omega - \mathbf{H}^-)^{-1} \right]_{\nu\mu} \langle \Phi_\mu | e^{-\sigma} a_p e^\sigma | \Psi_0 \rangle, \end{aligned} \quad (33)$$

with $\mathbf{H}^+$, $\mathbf{H}^-$ denoting the EOM Hamiltonian in the complete $N \pm 1$ excitation basis. Note that no coupling between these two sectors occurs due to Eq. (13). Furthermore, Φ_ν denotes excited Slater determinants in the $N \pm 1$ sectors [compare Eq. (17)]. Next, we write Eq. (33) compactly as

$$\mathbf{G}_{11}(\omega) = \begin{pmatrix} \mathbf{M}_{11}^\dagger & \mathbf{M}_{21}^\dagger \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{M}_{11} \\ \mathbf{M}_{21} \end{pmatrix} \quad (34)$$

(note that in the blocks the $N+1$ and $N-1$ are still decoupled). $\mathbf{M}_{11}$ denotes contributions from $\langle \Phi_\mu | e^{-\sigma} a_q^\dagger e^\sigma | \Psi_0 \rangle$ and $\langle \Phi_\mu | e^{-\sigma} a_p e^\sigma | \Psi_0 \rangle$ in the 1h-1p sector and $\mathbf{M}_{12}$ in all higher excitation sectors. It becomes clear that the terms such as $\langle 0 | e^{-\sigma} a_p e^\sigma | \Phi_\nu \rangle$ in Eq. (33) prohibit a simple projection of the decoupled EOM Hamiltonians $\mathbf{H}^+/\mathbf{H}^-$ in the 1-h/1-p space as in Eq. (29). Consequently the precise connection to the Dyson equation can be only made approximately by separating $e^{-\sigma} a_q^\dagger e^\sigma$ into zeros order term a_q and higher order terms of the BCH. Consequently, Eq. (34) is rearranged to

$$\begin{aligned} \mathbf{G}_{11}(\omega) &= \begin{pmatrix} \mathbf{1}_{11} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}_{11} \\ \mathbf{0} \end{pmatrix} \\ &+ \begin{pmatrix} \tilde{\mathbf{M}}_{11}^\dagger & \mathbf{M}_{21}^\dagger \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \tilde{\mathbf{M}}_{11} \\ \mathbf{M}_{21} \end{pmatrix}. \end{aligned} \quad (35)$$

From the above considerations for the one-particle Green's function it becomes apparent that a ground-state p/h transformation of the Hamiltonian $\hat{H}$ prohibits the determination of $\mathbf{G}$ through the Dyson equation [Eq. (7)] exactly. Instead, the different components of $\mathbf{G}$ have to be determined separately. (See also the work by Tarantelli and Cederbaum¹).

III. HEDIN VS PARQUET

REFERENCES

¹A. Tarantelli and L. Cederbaum, "Configuration-interaction formulation of the dyson equation," Phys. Rev. A **45**, 2790 (1992).